

COMPUTER ARCHITECTURE

PROJECT REPORT

ARCHITECTURE-LEVEL ENCRYPTION ENGINE
USING X86 ASSEMBLY



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INTRODUCTION



Data security is very important in modern computing systems. Encryption is used to protect information by converting readable text into an unreadable form.

This project not only performs encryption and decryption, but also demonstrates how instructions are processed inside the CPU using registers, memory addressing, and ALU operations. The system provides insight into the Fetch-Decode-Execute cycle of a processor and shows how data moves between memory and registers during execution.

PROBLEM STATEMENT

Most encryption programs are written in high-level languages. In such languages, the internal working of the CPU is hidden. Because of this, students find it difficult to understand how data is actually processed inside the computer.

This project solves the problem by implementing encryption using basic assembly language instructions and CPU registers.

OBJECTIVES

The main objectives of this project are:

- To understand CPU instruction execution
- To analyze register-level data processing
- To study memory addressing in Assembly
- To simulate ALU-based operations
- To observe instruction flow in processor architecture
- To implement encryption using low-level hardware concepts
- To analyze execution efficiency and instruction count

SCOPE OF THE PROJECT

This project is designed for learning purposes only. It focuses on:

Text encryption and decryption
Single-key encryption method
Character-by-character processing

It is not intended for real-world security systems.



TOOLS AND TECHNOLOGIES

ITEM	DESCRIPTION
Language	Assembly
Assembler	MASM 615
Library	Irvine32.inc
IDE	VS Code
Platform	Windows

ENCRYPTION TECHNIQUE

This project uses the XOR encryption technique.
How XOR Works:

- If two bits are same → result is 0
- If two bits are different → result is 1

Formula:

- Encrypted Text = Plain Text XOR Key
- Decrypted Text = Encrypted Text XOR Key
-

The same operation is used for both encryption and decryption.

METHODOLOGY

- 1.PROGRAM STARTS
- 2.MENU IS DISPLAYED
- 3.USER SELECTS AN OPTION
- 4.USER ENTERS TEXT
- 5.USER ENTERS ENCRYPTION KEY
- 6.PROGRAM APPLIES XOR OPERATION
- 7.RESULT IS DISPLAYED
- 8.PROGRAM RETURNS TO MENU
- 9.PROGRAM ENDS WHEN USER SELECTS EXIT

PROGRAM STRUCTURE

The program consistsof:

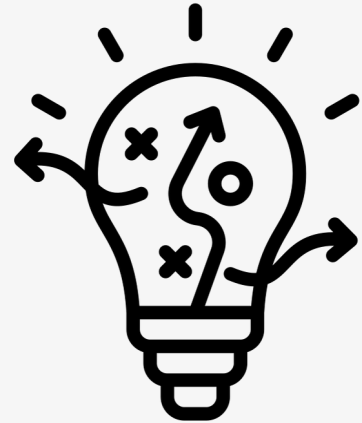
- Main procedure(menucontrol)
- Input section (text input)
- Key input section
- XOR processing procedure
- Output display section

CA CONCEPTS USED

CONCEPTS	USAGE IN PROJECT
Register	Store characters and keys
ALU	Performs XOR encryption
Memory Addressing	Accesses text buffers
Instruction Execution	CPU executes MOV/XOR instructions
Loops	Iterative character processing
Stack	Procedure calls and returns

ADVANTAGES

- Simple and easy to understand
- Strong COAL concept implementation
- Fast execution
- Menu-based interface



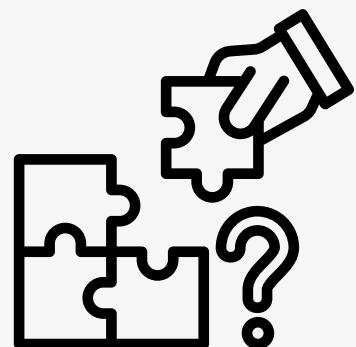
DISADVANTAGES

- Not suitable for real security
- Simple encryption method
- Single-byte key
- Limited text length



PROBLEMS FACED

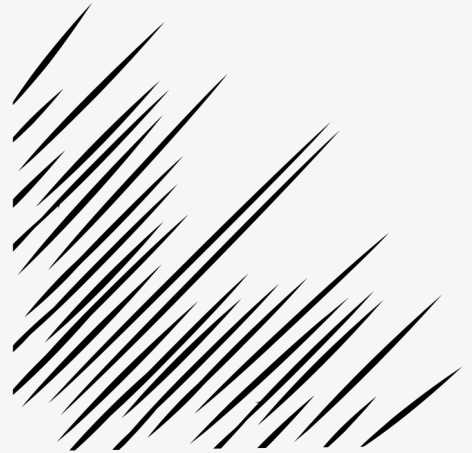
- MASM syntax errors
- Register size mismatch
- Loop handling issues
- String input handling



These were solved through debugging and testing.

LIMITATIONS

- Works only on Windows
- Supports only text input
- Fixed buffer size
- Same key required for decryption



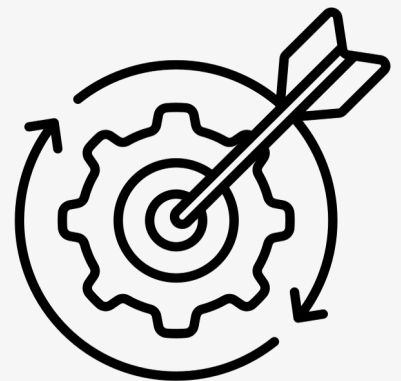
FUTURE IMPROVEMENTS

- The project can be improved by:
- Adding file encryption
- Using stronger encryption methods
- Supporting longer text
- Adding graphical interface
- Using multi-byte keys

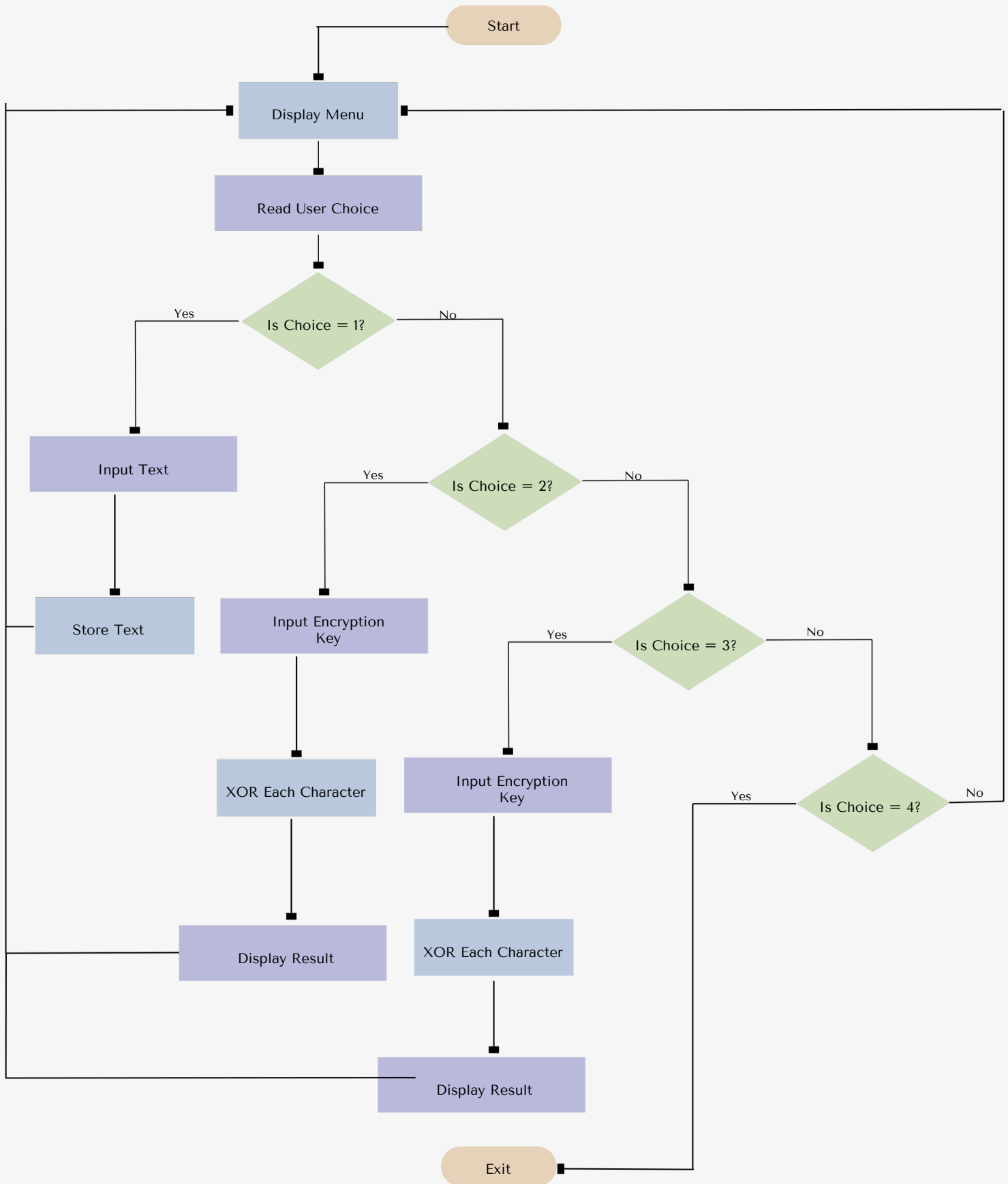
LEARNING OUTCOMES

From this project, the following were learned:

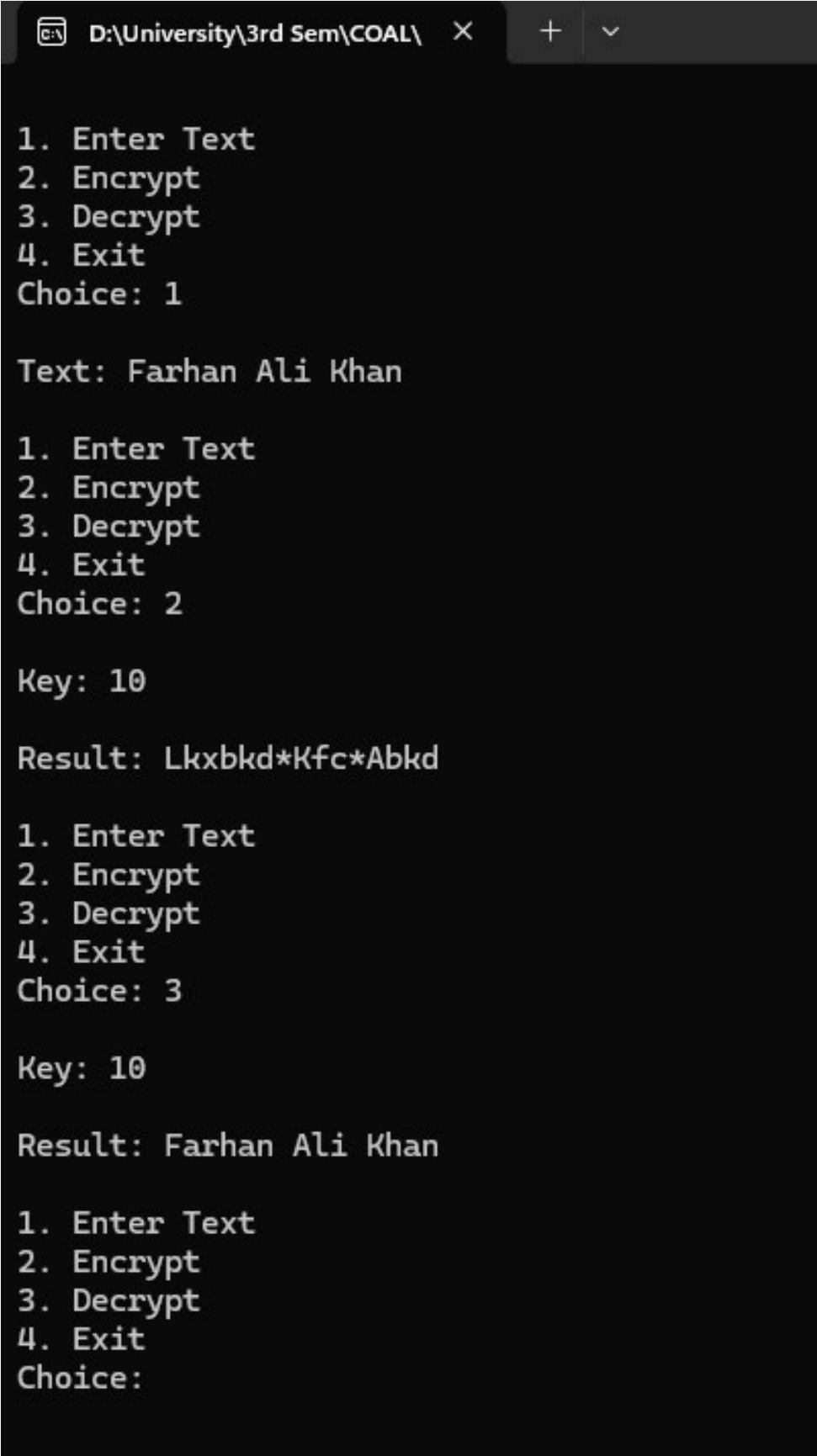
- Assembly language programming
- Working of CPU registers
- Memory usage
- Bitwise operations
- Program control using jumps and loops



FLOWCHART



CONSOLE OUTPUT



```
D:\University\3rd Sem\COAL\ X + v

1. Enter Text
2. Encrypt
3. Decrypt
4. Exit
Choice: 1

Text: Farhan Ali Khan

1. Enter Text
2. Encrypt
3. Decrypt
4. Exit
Choice: 2

Key: 10

Result: Lkxbkd*Kfc*Abkd

1. Enter Text
2. Encrypt
3. Decrypt
4. Exit
Choice: 3

Key: 10

Result: Farhan Ali Khan

1. Enter Text
2. Encrypt
3. Decrypt
4. Exit
Choice:
```

CONCLUSION

The Menu-Based Encryption and Decryption Tool successfully shows how encryption can be implemented using assembly language. The project helps in understanding how data is processed at the CPU level.

Although the encryption method is simple, it provides a strong foundation for learning computer organization and assembly language concepts.

REFERENCES

- KipR. Irvine–Assembly Language for x86 Processors
- MASM Documentation
- Intel x86 Manuals
- Computer ArchitectureCourse Notes

THANK YOU